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Beyond Single-Score Evaluation: A Bidirectional Analysis of Gender Bias in Language Models

Bachelor's Thesis (9 ECTS)

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Abstract:

Research on gender bias evaluation in masked language models has largely relied on single-direction frameworks, where gendered pronouns are masked and predicted from contexts such as occupations. Although this approach measures how models infer gender from context, such evaluation does not examine how gender influences the prediction of bias-related contexts. This thesis addresses this gap by investigating whether gender bias scores remain consistent across two evaluation directions: context-to-pronoun and pronoun-to-context. To support this investigation, two new bidirectional pronoun–context datasets are introduced, containing short and long sentences across three contexts: occupations, object nouns, and action verbs, with validation through human evaluation and annotation. Existing benchmarks are also adapted for bidirectional pronoun–context evaluation, and a framework is developed to measure gender bias in both directions and assess agreement and directional bias. Results across multiple benchmark datasets and BERT-family models show that agreement between the two directions varies by dataset and model, suggesting that single-direction evaluation may not fully capture gendered associations. These findings highlight bidirectional agreement as a more comprehensive approach to evaluating gender bias in masked language models.

Keywords: Artificial Intelligence, Language Models, BERT, Gender Bias

CERCS: P176 Artificial Intelligence

Kahesuunaline sooliste eelarvamuste hindamine keelemudelites.

Lühikokkuvõte:

Soolisi eelarvamusi keelemudelites on uuritud laialdaselt, kuid peamised sooliste eelarvamuste hindamise meetodid on ühesuunalised, mille puhul on võimalik, et need skoorid näitavad keelemudelites esinevaid soolisi eelarvamusi vaid osaliselt.

Selles bakalaureusetöös keskenduti soolistele eelarvamustele BERT perekonna maskeeritud keelemudelites. Töö eesmärk oli uurida soolisi eelarvamusi kahesuunalise meetodiga, mis mõõtis konteksti põhjal ja soostatud asesõna soolist eelarvamust ja vastupidiselt ka soostatud asesõna tekitatud soolist eelarvamust konteksti ennustamisel ning nende koondskoori. Selleks et eelarvamusi uurida, loodi kaks kontekstipõhist andmestikku ning lisaks sellele puhastati varasemalt avaldatud andmestikud.

Bakalaureusetöö tulemusena selgus, et kahesuunaline koondhinnang soostatud asesõna ja konteksti vahel varieerub olenevalt andmestikust ja BERT perekonna mudelist. Analüüs näitaski, et ühesuunaline sooliste eelarvamuste hindamine võib olla ebapiisav, et soolisi eelarvamusi hinnata maskeeritud keelemudelites ning oleks vaja kasutada kahesuunalist sooliste eelarvamuste hindamismeetodit.

Võtmesõnad: Soolised eelarvamused, keelemudelid, BERT, tehisintellekt

CERCS: P176 Tehisintellekt

Contents

1. Introduction:

Language models have become widely adopted across natural language processing (NLP) applications, demonstrating their capacity to perform a broad range of tasks. However, despite these advancements, previous research has consistently shown that these models exhibit biases [1]. These biases raise concerns about fairness and ethical deployment, as they can reinforce stereotypes and lead to discriminatory behaviour by models.

Transformer-based decoder models, such as large language models that power chatbots like ChatGPT [2], have gained widespread popularity. Meanwhile, BERT-family [3] language models, which utilise an encoder-transformer architecture, remain relevant, especially for delivering high-quality results on classification tasks [4]. Understanding the gender bias present in these models is crucial to preventing biased responses from the language models.

Previous work has studied learned social biases in language models, particularly in BERT-family models [5, 6]. However, most existing research relies on a single metric for bias evaluation using established benchmarks [1]. In addition, limited research has examined bias bidirectionally, for example, by considering both the relationship from occupational context to pronouns and from pronouns to occupational context. This thesis addresses this gap by investigating bias in both directions and measuring the degree of agreement between the resulting bias scores.

This thesis makes three main contributions. Firstly, two new datasets are created for evaluating language models. Then, existing benchmark datasets, including WinoBias [7], Winogender [8], and GenderLex [9], are adapted for bidirectional agreement evaluation. Third, a bidirectional evaluation framework is applied to measure the consistency between context-to-pronoun and pronoun-to-context bias scores. The results reveal that agreement varies across datasets and BERT-family models, indicating that unidirectional bias evaluation may be insufficient to fully capture the range of gendered associations encoded in bidirectional masked language models.

The thesis has the following structure. First, we describe stereotypes in language and society, then define bias in language models. Then, studies on social bias in language models are described. Third, the dataset creation used to evaluate gender bias in the thesis is outlined. Fourthly, the dataset creation used to evaluate gender bias in the thesis is outlined, including the description of data collection and construction, human annotation and validation, and the resulting datasets. Then, the evaluation chapter describes the bias measures and model selection used for evaluation. The experimental protocol is then outlined, followed by the results and discussion.

2. Background

This section will explain how the stereotypes in language are formed, define bias for the thesis, explore existing benchmark datasets, and previous evaluation methods that have been proposed on the topic. Then, the thesis's evaluation metrics are laid out.

2.1 Stereotypes in Language and Society

Stereotypes are pervasive social constructs that influence perceptions, behaviours, and even language, often reinforcing rigid norms about gender roles and identities. From an early age, individuals absorb these biases through cultural narratives, media, and everyday discourse, which can perpetuate inequalities and limit opportunities. Research has shown that such stereotypes not only shape societal expectations but also impact self-perception and interpersonal relationships [10]. They shape everyday interactions between people, what people think of what a womanly or manly job is, and which words refer to which gender, and these stereotypes can be embedded in the language people use.

According to the European Institute for Gender Equality (EIGE), which created the European Gender Equality Index, the European Union (EU) is at least 50 years away from achieving full gender equality between men and women. The EU average index sits at 63.4 out of 100, whereas Estonia's score is even lower, at 59.4. For example, in the *Power* section of the index, according to the 2025 index, the share of women on boards of the largest quoted companies in Estonia is only 14 % [11].

This shows that gender inequality remains an issue across the European Union as a whole, and even more so in Estonia. The Gender Equality Index, at the same time, does not take into account minorities, including gender minorities, and other marginalised groups in society, for which the inequality might be even higher than between the genders.

When looking at the workplace-related stereotypes and inequalities between men and women, according to McKinsey & Company's report *Women in the Workplace 2025*[1], which looks at the data from the United States, for every 100 men who are promoted to a manager role, only 91

[1] https://www.mckinsey.com/capabilities/people-and-organizational-performance/our-insights/women-in-the-workplace

women are promoted. The difference is even greater for women of colour: only 74 women of colour are promoted for every 100 men promoted to manager roles.

In addition, there is evidence that a more equal distribution of domestic and workplace-related labour between women and men is beneficial for both men, who are often seen as the *breadwinners* of the family, and women who are expected to do most of the domestic labour [12].

According to a 2025 study, both social knowledge and gendered linguistic cues increased gender-based bias in responses from Estonian-speaking participants (a grammatically genderless language) and from Russian-speaking participants (a grammatically gendered language). This highlights the importance of both learned social biases and linguistic biases in shaping people's biased views [13].

This shows that understanding social biases is crucial, as such biases may be encoded in language models during training. Investigating these biases is therefore important for mitigating their potential effects on the people who use these models.

2.2 Bias Definition

In this thesis, we draw inspiration for our definition of bias from the work of Gallegos *et al.* [1]. We view social bias not only as a statistical imbalance in the models, but also as unequal treatment or outcomes shaped by structural and historical power differences. In detail, we focus on representational harm, which includes misrepresenting social groups, promoting views that focus on dominant groups while dismissing others, or assigning rigid social roles based on group identity. In the context of large language model biases embedded in the training data and retained after training, can cause representational harm if one gender is preferred to another based on the surrounding context, or one biased context is preferred to another based on the pronoun [14].

2.3 Benchmark Datasets in Gender Bias

Gender bias in language models has been widely studied through benchmark datasets designed to evaluate stereotypical associations in model predictions [1]. There are two approaches to creating a dataset to evaluate social biases. The first option is to use crowdsourced sentences, which are structurally diverse and often created by humans, as in the CrowsPairs [5] dataset and the Setroset datasets [15]. The second option is to use template-based datasets such as WinoBias [7], which can be created either by humans or by large language models.

We decided to follow template-based datasets such as WinoBias [7], Winogender [8], and GenderLex [9] to provide a controlled environment for bias evaluation. Namely, the template-based approach ensures that sentences, in terms of structure, remain similar across the dataset, unlike crowdsourced datasets, which can contain too many structurally diverse sentences. The template-based approach aligns more closely with the thesis's approach, which evaluates biases in specific word groups, for which a controlled dataset structure is beneficial.

Template-based sentences follow Winograd-type schemes: puzzle-like instances in which a human or a computer must decide, in an English sentence, which personal pronoun matches the person previously referred to. While these examples are generally easy for humans to solve, they are difficult for computational models that, among other things, use gender bias in the data to solve them. Winogender [8] used Winograd-type sentences to demonstrate the gender bias of computational models, in which a person's job was an important component. The sentences they created contained three references to people. The first reference was to a person's job, for example, the *secretary*, the second reference was to another person, for example, *the passenger* or *someone*. The third reference was again to either the first or second person, depending on the type of sentence, using one of the English gendered pronouns (he/she/they). The models had to determine which of the two people the pronoun referred to, whereas the reference to the person did not depend on the pronoun's gender and was easy for humans to guess. For example, the sentence *The paramedic performed CPR on the passenger even though she/he/they knew it was too late.* was used, where it is immediately clear to an English speaker that regardless of the choice of pronoun, the pronoun refers to *the paramedic*. Testing three different models, they found that all used gender bias to solve these instances, leading to incorrect results. Removing gender bias from these models is important to ensure they do not give users false information.

Prior to Winogender, the WinoBias dataset [7] studied how computational models, solve Winograd-type sentences used over three thousand instances. Their sentences were also divided into two types and were similar to the sentences used in Winogender's study. In the first type of sentences, the pronoun referred to the first person of the two people referenced, and in the second type, the pronoun referred to the second person mentioned.

The current study also draws on a recent study, GenderLex dataset [9], on the same topic, which creates a synthetic dataset using LLMs to evaluate the impact of action verbs and object nouns on the gender bias in these models. They used a template-based sentence structure so that there are only two references to a person: one with the person's occupation (e.g., *the chef*) and the other with a pronoun (she/he). Furthermore, the sentence contains one noun (e.g., *the recipe*), and an action verb (e.g., *crafted*). To measure the gender biases introduced by action verbs and nouns in large language models within sentences, they evaluated the models' pronoun choices when the entire context was provided, with equal word weights or with the verb or the noun weighted more heavily. The study revealed that emphasising verbs increased gender bias towards the male pronoun *he*, while emphasising object nouns increased gender bias towards the female pronoun *she* [9].

2.4 Bias Evaluation in Language Models

Token prediction transformer decoder architecture, large language models, and BERT encoder masked language models are fundamentally different. In decoder models, a token within a sentence is predicted using all the previous tokens. In contrast, a token in a masked language model is predicted using all previous tokens and all tokens after the masked token [16]. For decoder models, therefore, if a set of tokens preceding the token to be predicted are $W_{<}t = (w_1, ...w_{t-1})$ then the logarithmic probability for a sentence W can be calculated easily by using the chain rule of probabilities, where the log probabilities of each token based on the preceding tokens are summed.

However, this left-to-right method is not directly applicable to encoder-based language models like BERT-family models, because the masked token predictions are done using $W_{\setminus t} = (w_1, ..., w_{t-1}, w_{t+1}, ..., w_{|W|})$. To address the issue, Salazar *et al* [16] proposed a method for pseudo-log-likelihood scoring of Masked Language Models (MLM). The logarithm of the probability is taken by masking all the tokens in the sentence one by one and calculating the log probability of the masked token:

$$PLL(W) = \sum_{t=1}^{|W|} \log P_S(w_t \mid W_{\setminus t}; \theta) \quad (1)$$

where $W = (w_1, \ldots, w_{|W|})$ is the input sentence, w_t is the token at position t, and $W_{\setminus t}$ is the sentence with w_t masked. $P_S(w_t \mid W_{\setminus t}; \theta)$ denotes the probability assigned to the masked token by the scoring model S, parameterized by the learned weights θ.

This scoring approach was later adapted to measure social biases with the CrowS-Pairs score for BERT-family language models, which is tailored to a dataset containing 1508 sentences [5]. More specifically, to score each sentence for biases, they used two types of words in each sentence. For example, for the sentence pair *[John] ran into his old football friend* vs *[Shaniqua] ran into her old football friend*, the modified words are the name and the gendered pronoun, and the unmodified words are all the remaining words in the sentence. For scoring, they masked all unmodified words one by one, calculating the unmodified tokens' log-likelihood score based on the PLL conditioned on the modified tokens, which were not masked in any iteration of the scoring. If the unmodified tokens are U and the modified tokens are M, then the equation for the scoring is:

$$\text{score}(S) = \sum_{u_i \in U} \log P(u_i \mid U_{\setminus u_i}, M, \theta) \tag{2}$$

Later, Kauf *et al.* [17] proposed a better approach to MLM scoring. The problem with the original PLL score, according to their study, is that longer and less frequent words get tokenised as multiple tokens; therefore, when the PLL scoring masks one token at a time, there is a data leak as the model can see both the previous and the following tokens from the masked token.

To address that, they proposed two new ways to do MLM scoring. The first approach was to mask all of the tokens that the word is tokenised into at once, so that the model sees none of the information about the word (target token). This approach is referred to as PLL_{ww}. The second method was to mask each following token from the same word as the current token, but not mask the tokens that preceded the current token from the same word; they refer to this approach PLL_{l2r}. They found that the second approach was more beneficial in terms of the score characteristics for scoring MLMs.

In this thesis, we take inspiration for the bias evaluation from the PLL score and the improved PLL_{l2r} score, which uses a within-word left-to-right ($l2r$) pseudo-log-likelihood scoring method. Specifically, when scoring the i-th subword of the target word w, we mask not only t_i but also all later subwords of w, while leaving earlier subwords of w and the surrounding sentence context visible.

Initially, we considered only the original PLL score for the evaluation. However, because our dataset relies on words that are tokenised into multiple subwords (e.g., occupations), we follow the method proposed in [17] and adopt the improved PLL_{l2r} score for the main evaluation, while also reporting a comparison with the original PLL score.

In particular, for the modified version of the score, we evaluate the log probability of each subword token of the target word w such that all later subwords of the same word are masked, while earlier subwords of w and the surrounding sentence context remain visible. Using the same notation as before:

$$\mathrm{PLL}(w)_{\mathrm{l2r}} \;=\; \sum_{i=1}^{|w|} \log P\Big(t_i \mid W_{\setminus\{t_i,\, t_{i+1},\, \ldots,\, t_{|w|}\}}\Big)\,, \tag{3}$$

where $W_{\setminus\{t_i,\ldots,t_{|w|}\}}$ denotes the sentence W with the current token t_i and all following subwords of the same word w replaced by `[MASK]`.

For each minimal pair (s_m, s_w) we report the signed preference $\Delta = \mathrm{PLL}_{l2r}(s_m) - \mathrm{PLL}_{l2r}(s_w)$, which serves as the unit of analysis for all subsequent per-context aggregation, where we calculate the aggregate score for each context type for each dataset per model, which we call the mean Δ score, which we call the context mean score for short. Then, to calculate the female-leaning and male-leaning percentages, we use the sigmoid function, where a negative mean delta indicates a leaning towards female bias, and a positive mean delta indicates a leaning towards male bias.

3. Data

Existing benchmark datasets for evaluating gender bias mainly focus on occupations, and there is limited work that extends beyond this scope [1]. In this thesis, previous work is expanded beyond occupational titles to include object nouns and action verbs. To do that, inspired by Genderlex [9], and to measure the model's sensitivity to sentence length, two datasets were created: one containing shorter sentences and another containing longer sentences matched in length to those used in previous research. The datasets were designed to include a diverse range of object nouns and action verbs, so that the results of the gender bias analysis would be generalisable beyond the datasets.

3.1 Data Collection

Based on previous work with the gender bias benchmark dataset WinoBias [7], we focus on the most common 40 occupational title biases[2] (e.g., *CEO* with males and *cleaner* with females), supported by the US labour force statistics. Therefore, both the long and short datasets followed a sentence structure in which the occupation was presented first. Optionally, there was some context, and the third part of the sentence was the action verb followed by optional context; the fourth part was the object noun followed by optional context; and last in the sentence was the gendered pronoun, which was either a male or female pronoun.

To generate short and later long sentences for each occupation that would follow the defined structure, triplets that included the **occupation**, an **action verb**, and an **object noun** were created. To do that, LLM[3] is used for keyword generation by giving 10 occupations at a time, in alphabetical order, along with the specific prompt, as can be seen in the Appendix (9.1.1). An example of a triplet generated would be: (laborer; shovel; dug), or (receptionist; phone; answered).

3.1.1 Methodology

After generating keyword extractions (i.e., occupation, verb, and noun), we rely on an iterative refinement process involving human quality assessment, as shown in Figure 1. For each of the datasets, the *context* parts were optional as long as the full sentence met the length requirement.

[2] https://uclanlp.github.io/corefBias/overview

[3] Claude Opus 4.6 model

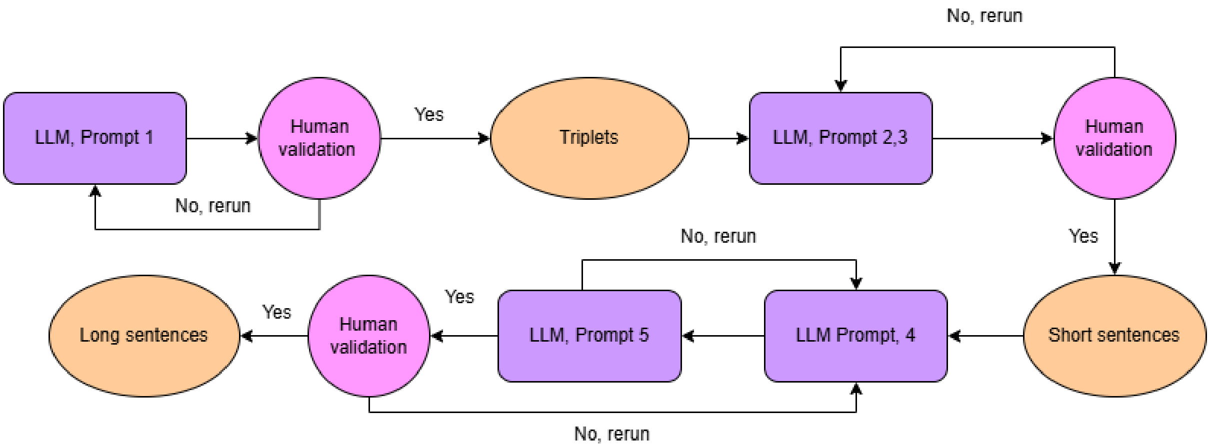

Figure 1. The pipeline of the datasets' creation. The triplets consist of an occupation, object noun, and action verb and the prompts for the dataset's creations can be further seen in the Appendix 9.1.

For both datasets, a zero-shot approach using the GPT-5 model[4] was used. No thinking modes were used. LLM was chosen to generate the dataset because writing high-quality, meaningful sentences by hand would have taken a long time or cost a significant amount. Therefore, using LLM to generate initial sentences and manually checking them is both cheaper and less labour-intensive.

3.1.2 Short Sentence Structure

The sentences to be created in the dataset followed the format, where the first word of the sentence mentioned the occupation, then, after optional additional context, the action verb was mentioned followed by the object noun after optional context, followed by the pronoun, which came last in the sentence. The short dataset was generated using the 400 unique triplets outlined previously (3.1). The sentences were generated one by one using the prompts and JSON schema outlined in the Appendix (9.1.2). This approach yielded 400 unique sentences that correctly followed the previously outlined sentence structure. The data were then manually checked for issues with sentence structure and meaning; for example, some sentences had an illogical order of words, which did not sound natural for an English speaker. Most of the sentences generated by the model were meaningful given the context. One important change in the sentences, in addition to some edits to the meaning, was removing special characters entirely.

[4]The model `gpt-5-2025-08-07`

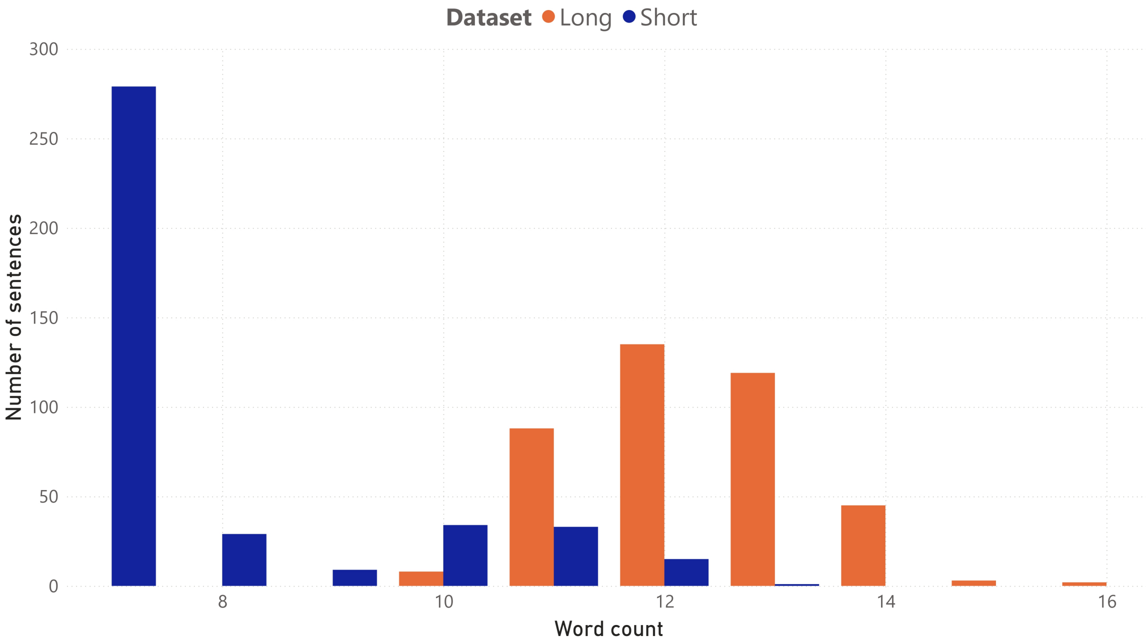

Figure 2. The distribution length of the shorter and longer dataset.

3.1.3 Long Sentence Structure

The main idea of generating shorter sentences is to validate the model bias in a shorter context and to see the sensitivity of the models based on sentence length. Therefore, after creating the dataset of short sentences, another dataset with the same sentence structure and sentence triplets was created. The longer dataset was created again using the GPT-5 model. The initial longer sentences were generated using the sentence lengthening prompt, as shown in the Appendix (9.1.3), which was then submitted to the sentence checker GPT-5 with a zero-shot prompt, which can be seen in the Appendix. The idea behind having two zero-shot prompts, one for generation and one for validation, was to reduce the number of sentences that did not meet the length requirements for the extended sentences. The model was instructed to use the short sentence as input and lengthen it with meaningful context, while keeping all the words from the shorter dataset and preserving the original sentence's word order. This was done to keep the shorter and longer datasets similar enough so that the biased results would be comparable between the two datasets. After creation, the dataset was thoroughly checked and cleaned of any special characters. The generated dataset had better context quality than the shorter dataset. However, a common mistake the GPT-5 model made was adding time-specific context, which is noise for the model when measuring bias. Table 1 shows the generated dataset with the triplets for short and long sentence structure datasets. Also, Figure 2 shows the difference in the distributions of the

Table 1. Examples of the generated dataset long and short with triplets context.

Context	Short	Long
Noun	The driver buckled the [**seatbelt**] before departure for [**him****her**].	The driver buckled the [**seatbelt**] before departure in the rear seat for [**him****her**].
Verb	The driver [**buckled**] the seatbelt before departure for [**him****her**].	The driver [**buckled**] the seatbelt before departure in the rear seat for [**him****her**].
Occupation	The [**driver**] buckled the seatbelt before departure for [**him****her**].	The [**driver**] buckled the seatbelt before departure in the rear seat for [**him****her**].

datasets. The shorter sentences are largely only seven words long, and only a few sentences are longer than that, whereas the longer sentences have a wider distribution around the twelve-word sentence length mark.

3.2 Human Annotation

The occupations were annotated by two annotators, who had to mark each occupation as a woman's (F) job or a man's (M) job. This was done to assess agreement using Cohen's Kappa [18] score between humans as a baseline for comparison with LLM-generated results. To calculate the Kappa score for a binary classification task, which was done in the thesis as the annotators were asked to mark the occupations or sentences as either male-leaning or female-leaning sentences, the first step was to calculate the observed agreement between the annotators. Then, the probability that the annotators agreed on both classes by chance is calculated, and the final equation for calculating the Kappa score is:

$$k = \frac{p_o - p_e}{1 - p_e}$$

For the thesis's datasets, the Kappa score based on the 40 occupations was used to indicate the score for both the long and the short datasets, because in the 400 sentences each occupation appears 10 times. The assumption is that the bias would be mainly influenced by the occupation in the sentence. An overview of the scores is provided in Table 2, with annotations for Kappa scores and agreements for each dataset.

An important part of this research was annotating the previous GenderLex [9], WinoBias [7], and Winogender [8] research datasets with at least two annotators to compare annotation agreement with the BERT models' resulting agreements, as shown in Table 3.

Table 2. The Kappa scores, observed agreements between the annotators. The thesis's datasets use annotations based on human annotations of 40 occupations from the WinoBias dataset.

Dataset	Kappa score	Observed agreement	Meaning
WinoBias [7]	0.83	0.92	Almost perfect agreement
Winogender [8]	0.41	0.71	Moderate agreement
GenderLex [9]	0.55	0.78	Moderate agreement
Thesis's datasets	0.65	0.83	Substantial agreement

Table 3. Agreement scores (i.e., human bias) between human subjects and model pronoun bias in WinoBias occupation context category.

	BERT	RoBERTa	ModernBERT	DistilBERT	DistilRoBERTa
Annotator 1	0.524	0.541	0.521	0.518	0.521
Annotator 2	0.537	0.546	0.521	0.523	0.524
Average	0.530	0.543	0.521	0.520	0.523

As mentioned previously, to measure the LM's bias across occupations, object nouns, and action verbs, 40 occupations were selected to generate sentences in both the long and short datasets, based on those used in the WinoBias benchmark dataset [7]. The annotations for female and male bias were added to the WinoBias dataset based on the percentage of women whose occupation names were among the 40 occupations, compared with the percentage of men with the same job. The percentages were taken from data published by the US Bureau of Labour Statistics [5], which were available at the time of writing the WinoBias dataset (2018). For each of these occupations, the percentages for female and male bias were updated using the newly released US labour stats dataset for 2024. 38 of the occupations used in 2018 were updated. For two of them, the 2018 annotations were used because the job titles for those occupations (cleaner, labourer) did not appear in the newly released dataset. The occupations with a bias towards male or female, and

[5] https://www.bls.gov/cps/cpsaat11.htm

the corresponding label, can be seen in the table in GitHub [6]. If there were multiple matching job titles, the resulting gender distribution percentage was calculated using the weighted mean of the different matching job titles. In the end, on average, the occupation distribution of women and men working in a role had shifted towards gender equality. On average, the 38 occupations had shifted 0.65 percentage points towards equality, and out of the 38 occupations, 21 had a positive shift towards equal distribution.

3.3 Validating Data

The datasets created alongside the datasets from the previous research were thoroughly validated. Firstly, to standardise the format of all datasets, punctuation marks were removed from each dataset. Each dataset was then checked for correctness. It turned out that multiple sentences in the longer dataset contained multiple differently gendered pronouns within a single sentence, making 5 out of 400 sentences invalid. There were also issues with the dataset, where incorrect pronouns like *herr* or *him* appeared in sentences that were supposed to use *her* as the context. In addition to the changes to the datasets created in this thesis, multiple edits were made to the datasets from previous works. The main challenge in cleaning the datasets was that the language models used to generate them produced sentences that, as a whole, were not similar to how a human would speak; for example, switching word order to match the required format, but making the sentences sound unnatural. It was a challenge to correct those sentences, and often the author needed to rewrite the whole sentence to both match the required sentence structure and sound natural.

3.4 Resulting Datasets

As a result of the dataset creation, validation, and cleaning, five datasets that were ready to be assessed for gender bias were created. This included modifying three previous datasets for the purpose of evaluating bias in GenderLex [9], WinoBias [7], and Winogender [8]. For the WinoBias dataset, only the sentences that were of type 2 were chosen, and for Winogender all the sentences, which had both an occupation and an action verb in the sentences. In turn, the size of the chosen subsets of these datasets can be seen in Table 4.

[6] https://github.com/sadesystem/Bidirectional_Bias_Thesis

Table 4. Resulting datasets overview with the dataset size, basic statistics and the number of unique action verbs and object nouns per dataset

Dataset	Size	Avg. Length	Median	Std. Dev.	Obj. Nouns	Act. Verbs
Winogender [8]	178	14.81	15	2.91	×	55
WinoBias [7]	785	12.55	12	1.98	×	187
GenderLex [9]	837	11.95	12	0.90	223	111
Thesis's short	400	7.91	7	1.57	359	283
Thesis's long	400	12.30	12	1.05	359	283

As a result of the keyword generation 400, unique (object noun; action verb; occupation) triplets were generated. The triplets have 40 unique occupations, each of which repeats exactly 10 times, and across all triplets 359 unique object nouns, while only three object nouns repeat more than two times: *memo*, *chart*, and *phone*, each of them repeats three times. In addition, there are 283 unique action verbs; each repeats at most 4 times. This means the dataset has a very diverse set of triplets, which underpin the short and long datasets, making the results derived from it more generalisable beyond the data. The dictionaries with triplets can be found in the GitHub repository[7] of the thesis. All of the triplets were verified and checked by the author of this thesis.

The overview of the resulting datasets can be seen in Table 4. For the thesis's shorter dataset, the resulting dataset had sentences with a mean length of 7.91 and a median length of 7 words. Two versions of each sentence were created, one where the ending pronoun was masculine he/him/his, and one where the ending pronoun was she/her/hers. For the longer dataset, the resulting dataset was 12.30 words on average, with a mean length of 12 words. The length of the longer dataset is comparable to the WinoBias [7], which has average length of 12.5 words and median length of 12 words, GenderLex [9] with a mean of 12.0 words and median of 12 words and Winogender [8] with a mean of 14.81 words and median of 15.0 words. The comparison between the distributions of sentence lengths in the thesis's shorter and longer datasets, in words, is shown in Figure 2.

[7] https://github.com/sadesystem/Bidirectional_Bias_Thesis

As a result of the data creation, datasets in the following format were generated. Each dataset included the male pronoun version of the sentence, the female pronoun version, and the context to be masked. The sentences begin with the occupation, followed by the action verb, object noun and gendered pronoun at the end of the sentence. This sentence structure differs from that in previous research: in the WinoBias [7] dataset, two occupations were mentioned in a single sentence, and in the Winogender [8] dataset, two gendered pronouns were in each sentence. In addition, the Genderlex [9] dataset had the reversed order of object nouns and action verbs in the sentence. As a result of the data cleaning and creation, 11 datasets were created. For each sentence with a male pronoun, an identical sentence with only the pronoun changed to a female pronoun was created. Some sentences in the WinoBias and Winogender datasets were corrected. For example, in the sentence: *The person bought the guard a gift and wish him had a nice holiday* the word *wish* was changed to *wished* as this makes more sense within the sentence.

4. Evaluation

The current thesis uses modified versions of the original PLL score: $PLL(w)$ and $PLL(w)_{l2r}$, which were described in the Chapter 2.4. For each context category (occupation, object noun, action verb), the $PLL(w)_{l2r}$ score is computed on both the male pronoun and female pronoun sentence, for which the delta score Δ is then calculated, which is the $PLL(w)_{l2r}$ for the female sentence substracted from the $PLL(w)_{l2r}$ score of the male sentence. The mean of the delta per model, dataset, and pronoun is then calculated, indicating female-leaning bias with a negative score and male-leaning bias with a positive score.The scores are used to predict the gendered pronoun based on context, and the context is then used to predict the gendered pronoun in five different BERT-family models. In contrast, previous research has examined one-directional bias, such as in the CrowS-Pairs study with BERT-family models [5]. The main score we use for evaluation is $PLL(w)_{l2r}$, as this is, according to recent research [17], a better scoring method for masked language model scoring, but a comparison between the different model scoring methods is also laid out.

4.1 Model selection

Five models were selected for evaluation, all of which are widely used publicly available masked language models: BERT[8] [3], RoBERTa[9] [19], and recent model Modern-BERT[10] [4], and Distil model [20]: DistilBERT[11], and DistilRoBERTa [12].

4.2 Experimental Protocol

The experiment was done in two parts. Firstly, we masked the gendered pronoun in the sentence and asked the model to predict which gendered pronoun is under the mask. This way, we obtained log scores for the male-gendered pronoun (e.g., *him*) and the female pronoun (e.g., *her*). We then calculated the delta between the two scores, subtracting the female pronoun $PLL(w)_{l2r}$ score from the male pronoun score. The score was negative if the model preferred the female pronoun and positive if it preferred the male pronoun. To calculate the male and female probability percentage, the sigmoid function on the delta score was used.

[8] https://huggingface.co/google-bert/bert-large-uncased
[9] https://huggingface.co/FacebookAI/roberta-large
[10] https://huggingface.co/answerdotai/ModernBERT-large
[11] https://huggingface.co/distilbert/distilbert-base-uncased
[12] https://huggingface.co/distilbert/distilroberta-base

Secondly, we used the two gendered sentences that differed only in the gendered pronoun to calculate the score for the masked context token, which was either an occupation, an action verb, or an object noun. We followed the same procedure as for predicting gendered pronouns. First, the context was masked, as required for the $PLL(w)_{l2r}$ score, and the log-likelihood of the masked token was calculated. Because the model received sentences that differed only in the gendered pronoun, we could again calculate the bias towards female and male pronouns. Then, again, the context delta score equalled the female context score minus the male context score. The probabilities for male and female biases using the sigmoid function were calculated.

Lastly, the score we calculated was the agreement percentage, which showed the percentage of times the masked context aligned with the direction of the masked pronoun bias. A score near the 50% mark indicates that the bidirectional agreement of the bias is close to a random chance, which could indicate that the biases in one direction are significantly different from the other direction, or that the models struggle to give biased responses in one or both directions. A score significantly above the 50% mark would indicate that the bidirectional agreement is high for both models, and a score below the 50% mark would indicate that the models do not agree bidirectionally on the bias. However, they prefer different directions of bias based on the context.

4.3 Results

In this section, the results of the evaluation across context types, triplets, datasets and models are laid out, beginning with the context-level results for the thesis's datasets, followed by the analysis of the benchmark datasets' context-level biases and pronoun-level biases. Lastly, the two scoring methods are compared with each other.

Note that, for the main comparison with other benchmark datasets, the Winogender dataset was excluded because the adapted dataset was very small, having only 178 sentences, which makes it difficult to draw reliable conclusions about model bias. For the dataset with 10 bias categories in total (2 per model), only 2 showed statistically significant results, as shown in Table 9 for all dataset comparisons.

Thesis's Short and Long Datasets

As shown in Table 5, for the noun category of bias in the thesis's short dataset, all results were female-leaning across models, except for DistilRoBERTa. The bidirectional biases for the noun category were all close to random chance, which could indicate that the directions of biases for

Table 5. Context to pronoun preference on the noun, verb, and occupation subsets of the datasets used in this thesis. Asterisks denote significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

	Noun (%)				Verb (%)				Occupation (%)			
Model	M	F	mean	Agr	M	F	mean	Agr	M	F	mean	Agr
Thesis's short dataset												
BERT	34.8	65.2	-0.536^{***}	51.7	40.2	59.8	-0.192^{***}	52.2	41.5	58.5	-0.113^{***}	58.5
RoBERTa	37.8	62.2	-0.156^{***}	45.8	34.0	66.0	-0.252^{***}	44.0	45.5	54.5	-0.122^{**}	52.5
ModernBERT	45.0	55.0	-0.065^{*}	**54.2**	40.2	59.8	-0.126^{***}	**53.5**	48.5	51.5	-0.004	63.7
DistilBERT	41.5	58.5	-0.052^{**}	50.7	55.8	44.2	0.086^{***}	52.5	35.2	64.8	-0.181^{***}	48.0
DistilRoBERTa	55.2	44.8	0.030	52.5	50.8	49.2	-0.021	43.5	35.5	64.5	-0.279^{***}	**65.2**
Thesis's long dataset												
BERT	49.5	50.5	-0.010	47.0	42.2	57.8	-0.053^{*}	49.2	52.5	47.5	-0.006	48.0
RoBERTa	40.8	59.2	-0.105^{***}	48.0	33.8	66.2	-0.186^{***}	47.0	49.2	50.8	-0.059	57.0
ModernBERT	43.8	56.2	-0.070^{**}	**55.0**	40.5	59.5	-0.091^{***}	**55.8**	58.2	41.8	0.116^{***}	52.0
DistilBERT	45.0	55.0	-0.041^{***}	52.2	58.2	41.8	0.049^{***}	51.0	47.2	52.8	-0.082^{***}	58.5
DistilRoBERTa	48.8	51.2	-0.007	54.0	44.0	56.0	-0.016	44.8	39.2	60.8	-0.161^{***}	**66.5**

pronouns and object nouns differ, or that the models struggled to produce biased responses in the noun category. The average female-leaning percentage was 60%[13] and the average agreement was 51%. The strongest bidirectional agreement was observed for the ModernBERT model, with 54.2% agreement. For the action verb category, the biases were also female-leaning across all models except DistilRoBERTa and DistilBERT, averaging 57%. The average agreement was again close to random, averaging 49%. The occupation category showed the strongest agreement on bidirectional bias, with both ModernBERT and DistilRoBERTa showing bidirectional bias agreement of 63.7% and 65.2% respectively, and the overall bias agreement was 58%. This could be due to the occupation having a stronger bias embedded in the models in both the context and pronoun directions. The average female-leaning context preference was 59%. Overall, all the context categories displayed a female-leaning bias. In contrast, only the occupation category showed a higher bidirectional agreement with the bias, whereas the noun and verb categories showed bidirectional agreement close to random chance.

[13] All of the mean percentages for the object noun, action verb, and occupation categories are taken across statistically significant results.

Table 6. Context-to-pronoun preference on the GenderLex and WinoBias datasets. Asterisks denote significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

	Noun (%)				Verb (%)				Occupation (%)			
Model	M	F	mean	Agr	M	F	mean	Agr	M	F	mean	Agr
GenderLex dataset												
BERT	49.7	50.3	0.011	50.1	66.5	33.5	0.147***	55.4	63.9	36.1	0.156***	55.7
RoBERTa	49.9	50.1	−0.006	48.5	50.4	49.6	−0.026*	54.2	61.4	38.6	0.231***	69.3
ModernBERT	52.1	47.9	0.011	50.5	78.9	21.1	0.344***	38.8	80.8	19.2	0.415***	42.9
DistilBERT	48.3	51.7	−0.019*	53.3	87.8	12.2	0.628***	**80.4**	68.2	31.8	0.152***	**71.1**
DistilRoBERTa	53.2	46.8	0.010	**57.0**	79.1	20.9	0.275***	74.3	58.6	41.4	0.116***	66.8
WinoBias dataset												
BERT					50.8	49.2	0.006	52.4	74.1	25.9	0.721***	**79.2**
RoBERTa					52.7	47.3	0.018	54.5	60.3	39.7	0.292***	77.3
ModernBERT					49.7	50.3	−0.002	56.6	61.9	38.1	0.224***	76.3
DistilBERT					62.2	37.8	0.072***	**61.8**	69.2	30.8	0.170***	70.6
DistilRoBERTa					51.8	48.2	−0.026	52.9	61.7	38.3	0.271***	64.5

As shown in Table 5, in the thesis's longer dataset, the female leaning average for the noun category was 57%, which was lower than for the thesis's shorter dataset. The overall agreement was similar to that in the shorter dataset, averaging 48%. This might indicate that, even with a longer context for the model, the bidirectional bias remains significantly different across models. For the action verb category, the female-leaning bias was 56%, similar to the results from the thesis's shorter dataset. This might indicate that the bias in this category does not change significantly with added context. The overall agreement for models was similar to the previous results, averaging 50%. The occupation category bias shifted toward male bias with longer added context, which might indicate that if the context is too short, the true bias of the model is not as strongly measurable in the occupation category. The mean female-leaning context percentage was 52%, a 7 percentage-point difference compared to the shorter dataset. The overall bias agreement was 56%.

Benchmark Datasets

In Table 6, for the GenderLex dataset, all models displayed a significantly lower female-leaning bias in each context category than in both of the thesis's datasets. The mean female leaning percentage for object nouns, action verbs, and occupation categories was 51.7%, 27%, 33%,

Table 7. Example of match (√) or mismatch (×) with both bias direction for occupations context Bias_c and pronoun Bias_p for all dataset for ModernBERT.

WinoBias						GenderLex					
Context	Δ_c	Bias_c	Δ_p	Bias_p	Match	Context	Δ_c	Bias_c	Δ_p	Bias_p	Match
developer	1.817	M	1.665	M	√	psychologist	-0.031	F	-1.002	F	√
supervisor	-0.042	F	0.796	M	×	conductor	1.274	M	-1.277	F	×
mover	0.531	M	1.709	M	√	mathematician	0.065	M	0.899	M	√
writer	0.374	M	0.155	M	√	doctor	0.551	M	0.423	M	√
attendant	-0.254	F	0.267	M	×	leader	0.568	M	-1.948	F	×
Thesis's short						Thesis's long					
Context	Δ_c	Bias_c	Δ_p	Bias_p	Match	Context	Δ_c	Bias_c	Δ_p	Bias_p	Match
salesperson	-0.739	F	-0.463	F	√	receptionist	-0.019	F	-2.229	F	√
sewer	0.224	M	-0.702	F	×	auditor	0.314	M	-0.175	F	×
clerk	0.103	M	0.915	M	√	nurse	-0.331	F	-1.401	F	√
cook	-0.104	F	-0.963	F	√	carpenter	0.002	M	1.023	M	√
supervisor	0.080	M	-0.324	F	×	accountant	0.485	M	-0.833	F	×

respectively. The female-leaning bias for the object noun categories overall was 52%. In contrast, the action verb category was much more male-leaning in terms of bias compared to the thesis's short dataset, averaging 73%. The bidirectional agreement for the object noun category remained at around 50%. However, for the distilled models in the action verb category, the bidirectional bias agreement was much higher than for the models evaluated on the thesis's dataset, at 80.4% and 74.3% for the distilled BERT and RoBERTa models, respectively. ModernBERT, on the other hand, displayed low agreement of 38.8%. The occupation category, in addition to being high in male bias, was characterised by relatively high bias agreement scores, averaging 61%.

Also, in Table 6, for the WinoBias dataset, we measured only the biases and agreements in the action verb and occupation context categories. The action verb category showed a male-leaning bias, with the only significant result showing 62%, and bidirectional agreement was slightly above random guess level, averaging 56% across models. The occupation category, similarly to the GenderLex dataset, was significantly male-biased, averaging 65%. The bidirectional bias agreement score was also high, averaging 74%.

In Table 7, examples of the bidirectional disagreement calculation are shown using the ModernBERT model on both benchmark and our datasets. The disagreement score is calculated as the proportion of mismatches per context (occupation). One interesting observation is that Stereotype-aligned roles (*developer*, *nurse*, *receptionist*) agree in both directions, while ambiguous or leadership terms (*supervisor*, *auditor*, *leader*) disagree.

4.3.1 Pronoun Preferences

The pronoun preference scores in Table 8 can be divided into three parts. Firstly, the thesis's long dataset was the only dataset to have a slightly female-leaning pronoun bias with an average of 54% Secondly, the thesis's short dataset showed a varied female preference, with the highest at 71.5% and the lowest at 25%, averaging, perhaps, indicating that the shorter sentences in the dataset might not have provided enough context for the models to give biased responses. Thirdly, both the GenderLex and the WinoBias datasets showed a strong male pronoun preference, averaging 70% and 80%, respectively.

Table 8. Pronoun preference scores on bidirectional agreement on occupation context. The bidirectional agreement score is based on the WinoBias occupation category. Non-significant values (mean delta not different from zero at $p < 0.05$) are shown in gray.

	WinoBias (%)			GenderLex (%)			Thesis's long (%)			Thesis's short (%)		
Model	M	F	Agr	M	F	Agr	M	F	Agr	M	F	Agr
BERT	79.1	20.9	79.2	67.9	32.1	55.7	33.5	66.5	48.0	28.5	71.5	58.5
RoBERTa	73.9	26.1	77.3	79.0	21.0	69.3	53.2	46.8	57.0	75.0	25.0	52.5
ModernBERT	76.8	23.2	76.3	27.0	73.0	42.9	28.8	71.2	52.0	48.8	51.2	63.7
DistilBERT	91.5	8.5	70.6	89.2	10.8	71.1	73.8	26.2	58.5	68.8	31.2	48.0
DistilRoBERTa	80.9	19.1	64.5	87.8	12.2	66.8	47.8	52.2	66.5	54.2	45.8	65.2

4.3.2 Benchmark Dataset and Score Comparison

Table 9 shows an overview comparison of all datasets on the triplets context. Excluding the Winogender dataset, which only had two statistically significant out of the ten bias deltas, the models had the least statistically significant biased results (i.e., most results that could be explained by random chance) for the object noun category, where models displayed statistically significant biased results for only 8 out of 15 tests. The action verb category showed statistically significant bias in 14 out of 20 tests, and the largest number of biased results occurred in the occupation category, which showed 17 out of 20 tests with statistically significant bias.

Table 9. Overview of Context Bias mean delta values across all datasets and models (using $PLL(w)_{l2r}\Delta$ scores).

Dataset	Category	BERT	RoBERTa	ModernBERT	DistilBERT	DistilRoBERTa
WinoBias	Action Verb	0.006	0.018	-0.002	0.072^{***}	-0.026
	Occupation	0.721^{***}	0.292^{***}	0.224^{***}	0.170^{***}	0.271^{***}
WinoGender	Action Verb	0.029	0.011	0.016	0.015	0.008
	Occupation	0.405^{***}	0.096	0.077	0.038	-0.103^{**}
GenderLex	Action Verb	0.147^{***}	-0.026^{*}	0.344^{***}	0.628^{***}	0.275^{***}
	Object Noun	0.011	-0.006	0.011	-0.019^{*}	0.010
	Occupation	0.156^{***}	0.231^{***}	0.415^{***}	0.152^{***}	0.116^{***}
Thesis's short	Action Verb	-0.192^{***}	-0.252^{***}	-0.126^{***}	0.086^{***}	-0.021
	Object Noun	-0.536^{***}	-0.156^{***}	-0.065^{*}	-0.052^{**}	0.030
	Occupation	-0.113^{***}	-0.122^{**}	-0.004	-0.181^{***}	-0.279^{***}
Thesis's long	Action Verb	-0.053^{*}	-0.186^{***}	-0.091^{***}	0.049^{***}	-0.016
	Object Noun	-0.010	-0.105^{***}	-0.070^{**}	-0.041^{***}	-0.007
	Occupation	-0.006	-0.059	0.116^{***}	-0.082^{***}	-0.161^{***}

DistilBERT showed a biased result for 11 measurements, while BERT, RoBERTa, ModernBERT, and DistilRoBERTa had 7, 8, 8, and 5 significant results, respectively.

As can be further seen in the Appendix (9.2), the scores for the $PLL(w)_{l2r}$ (within_word) and $PLL(w)$ (original) differed very little across all socres, indicating that for the specific datasets and adjusted scores this thesis implements, the method of scoring does not change the overall bias evaluated for the models by much, although some of the differences between the resulting scores were statistically significant. For example, the differences for the original and within-word scores in the WinoBias dataset can be seen in Table 10. Across the board, the occupation category has the highest number of statistically significant differences between scores, which indicates that in this category, there were the most words that split into multiple tokens instead of consisting of one token, which changes the scores for the two different scoring methods. For example, the occupations *housekeeper* and *salesperson* could have been split into multiple tokens as they are longer and consist of two different words. In contrast, the action verb and object noun categories did not have as many statistically significant differences between the scoring methods, indicating that they were split into multiple tokens less often compared to the occupation category.

5. Discussion

The main finding of the analysis is that, especially in the object noun and action verb categories, bidirectional agreement is low in both datasets. This indicates that the framework of using bidirectional agreement categories compared to the previous one-directional bias assessment could be crucial for understanding the biases embedded in BERT-family language models and in mask language models in general. The highest bidirectional agreement across datasets for the models was in the occupation category, suggesting that occupation is more similarly biased bidirectionally. Still, the bidirectional agreement scores were not very high, ranging from $50 - 80\%$, indicating that, for the occupation-level bias assessment, a bidirectional approach would still be beneficial for uncovering bias patterns that unidirectional assessments do not reveal. A clear example is *attendant* in the WinoBias dataset, the context-based signal leans female while the pronoun-based signal leans male, producing a directional mismatch that a single-direction could not have detected.

Figure 3 shows model-level comparison per context. Across the board, the object noun category is the only female-leaning category, while the action verb category is moderately male-leaning, and the occupation category is the most male-leaning. Although the object noun and action verb categories in the thesis's datasets were similar in bias strength, the GenderLex object noun category was much more female-leaning than the GenderLex action verb category. In addition, the WinoBias dataset had slightly more male-leaning action-verb bias than the other model's object-noun bias. This finding of object nouns being across the board slightly more female-leaning compared to action verbs could be, firstly, explained by differences in the datasets, as we saw the biases for the object noun categories were close to being equal to a zero-level context delta, which might indicate that object nouns do not carry much bias in the models. This still does not explain the significant difference between the models in the GenderLex dataset. The difference might be due to features of the dataset that differ from those of the other datasets; for example, the order of action verbs and object nouns is reversed, which might influence the bias evaluation.

Nevertheless, if it is not the case that object nouns and action verbs are differently biased due to data or evaluation characteristics, and if, across the board, action verbs are still more male-leaning than object nouns due to biases in the models, then that could be explained by men being seen as more active agents. In contrast, the models portray women as more passive agents, which could explain differences in model preference across contexts: women are more closely associated

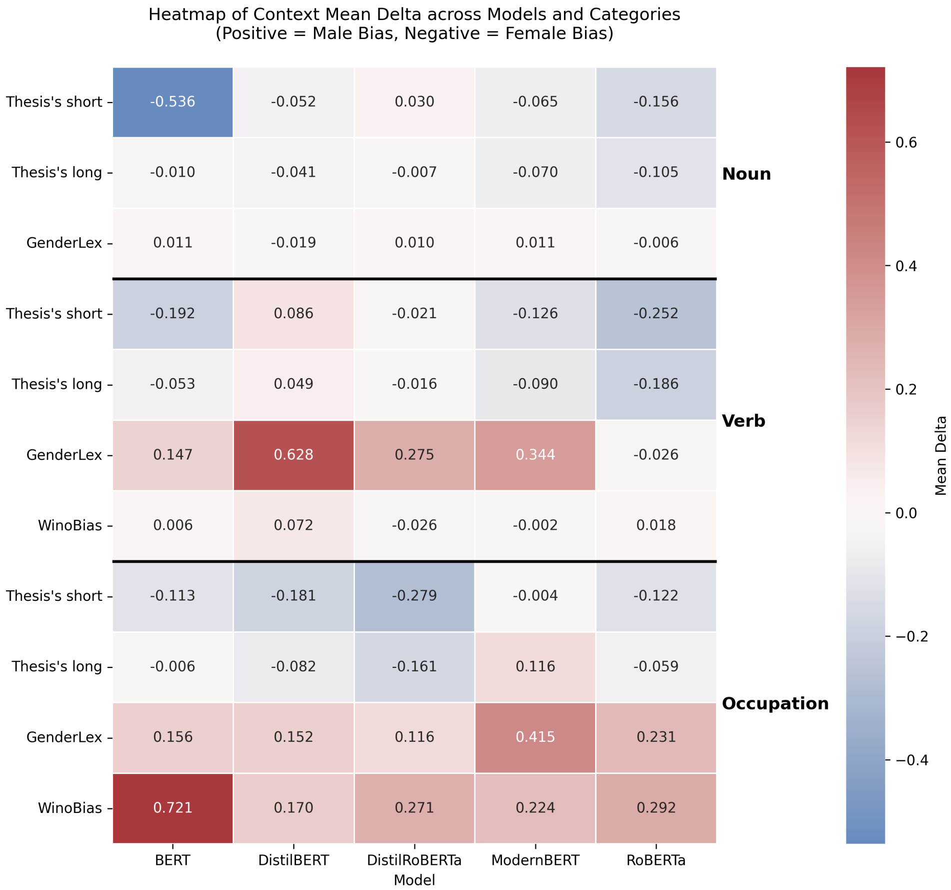

Figure 3. Overview of the results per category and model grouped by context type. The color grading indicates the mean delta; (+) indicates a bias towards male, (–) towards female.

with object nouns, while men are more closely associated with action verbs. This finding aligns with previous work in philosophy, where the famous feminist philosopher Simone de Beauvoir writes that: “Humanity is male, and man defines woman, not in herself, but in relation to himself; she is not considered an autonomous being. Woman, the relative being…” [21].

In addition, this reading is supported by Knowles [22], who interprets Beauvoir’s *The Second Sex* as follows: “Similarly, for Beauvoir, women’s complicity in their own subordination is expressed in the way women cling to limited and limiting self-conceptions that reduce their agency by casting them into the role of the ’Other’: man’s passive and dependent counterpart”. Furthermore, this is explained by the passivity-activity dichotomy within the femininity-masculinity binary, in which identifying as a passive object in relation to an active agent is seen as constitutive of a woman’s identity. In contrast, the active agent whose identity is defined in and by itself is the man’s, as Beauvoir seems to argue [21].

This finding is further supported by previous GenderLex datasets [9], which found that object nouns are more biased toward women and action verbs are more biased toward men across different GPT models. This possible interpretation of the results in the thesis should be further researched, using more data to verify the strength of the bias embedded in the language models.

Additionally, when we look at the statistically significant results for the action verb and object noun categories in the GenderLex dataset, the average bias towards men is 0.274 in the action verb category. At the same time, the object noun category has a female-leaning mean bias delta (Δ) of -0.019. At the same time, for the thesis's short dataset, the female-leaning mean bias Delta is -0.202 for the object noun category, whereas the action verb category is more male-biased, with a Delta of -0.121. For the thesis's long dataset, both scores are similar, with the action verb category being more female-leaning at -0.095, while the object noun category has a mean Delta of -0.072 across the statistically significant scores.

Also, the datasets used showed significantly different bias scores, suggesting that datasets can significantly affect the bias evaluation of different models. The thesis's datasets, which included a very diverse set of object nouns and action verbs, showed lower overall bias scores and a more female-leaning bias profile than the GenderLex and WinoBias datasets. This might be caused by a property which was laid out in the paper [17], where the thesis's scoring method was proposed, where they found that more frequent words in the training dataset have a higher PLL score and less frequent words have an overall lower PLL score, which would explain why a very diverse, and therefore possibly less frequent words in the training corpora of the models, lead to more female-leaning or neutral overall bias results. In addition, the results for the thesis's shorter dataset differed significantly from those in the longer datasets, suggesting that context length plays a significant role in biasing model responses.

Another observation is that comparing BERT with its distilled counterpart, DistilBERT, reveals a further trend in the context-bias breakdown across categories. For action verbs, BERT averages mean Δ of 0.131 (significant in 3/4 datasets), while DistilBERT averages 0.209 (4/4). For object nouns, BERT shows 0.536 but only in 1/3 datasets, while DistilBERT shows weaker but consistent biases of 0.037 (3/3). In occupations, BERT averages 0.330 (3/4), while DistilBERT averages 0.146 (4/4). Overall, BERT produces larger biases that are significant in fewer categories (7/11), while DistilBERT shows smaller biases that are significant in every category (11/11). However, the story is different for the pronoun-level bias, where both models displayed statistically significant bias across all datasets. However, the average mean deltas were higher for the DistilBERT model than for the BERT model, with averages of 0.997 and 0.692 respectively.

Another interesting finding is related to the comparison between the original BERT model and ModernBERT, which relies on recent advances in LLMs training, like rotary embeddings and longer context. For action verbs, ModernBERT averages an absolute significant Δ of 0.187 compared to BERT's 0.131 (both significant in 3 out of 4 datasets). For object nouns, ModernBERT averages 0.068 (2 out of 3 datasets) compared to BERT's 0.536 in a single dataset. In the occupation category, ModernBERT averages 0.252 compared to BERT's 0.330 (both in 3 out of 4 datasets). The bias scores for both models are male-leaning across all contexts, and the pronoun-level bias is also strong for both variants. This might indicate that the newer ModernBERT model, compared to the original BERT model, has still learned gender biases in a similar way the original model did, therefore even the bigger training data size [14] that the ModernBERT was trained on, displays similar biases compared to the smaller dataset[15] on which the BERT model trained on.

Finally, the pronoun-level biases were stronger than the scores with context-masking, which indicates that the models give biased responses more easily in the verb/noun/occupation-to-pronoun bias direction than in the pronoun-to-verb/noun/occupation direction. This could be explained by the fact that, if the pronoun is masked, there are fewer dictionary options likely to fill the gap. In contrast, if the verb/noun/occupation is masked, the possible set of words that could fill the gap is longer, based on the surrounding context, which could fit into the masked token.

[14]The training data was 2 trillion tokens

[15]3.3 billion words in total. $800M$ BookCorpus words and $2500M$ English Wikipedia words

6. Limitations

The work had multiple limitations. Firstly, the thesis's datasets, although diverse in terms of object nouns and action verbs, could have been longer, as 400 sentences might not yield meaningful and generalisable results. For example, the previous GenderLex dataset is more than twice as large, and the part of the benchmark WinoBias dataset used with type 2 sentences is also considerably longer.

In addition, the evaluation on the thesis's shorter dataset showed that the models lacked sufficient context and biased responses when masking the context or pronoun varied a lot due to that. Therefore, results from the shorter dataset might not reflect the models' actual bias.

Additionally, the thesis is based solely on English-language datasets, which could limit the generalisability of the results beyond the English language. In addition, the current research limits the model's responses to binary gender classification (male or female-leaning).

Lastly, the thesis used only three word types for the bidirectional bias evaluation: object nouns, action verbs, and occupations. Although these word types are popular in sentences, we could have evaluated biases on a new set of word types that previous research has not examined.

7. Conclusion and Future Work

In this thesis, we closed the gap by creating two diverse datasets of 400 sentences each, and by evaluating the language models with a novel bidirectional approach. In one direction, the context (action verb, object noun, occupation) is masked, and the score is calculated on the male-pronoun sentence and then on the female-pronoun sentence. In the opposite direction, the pronoun is masked and $\mathrm{PLL}(w)_{\mathrm{l2r}}$ is computed for both male-pronoun and female-pronoun versions. The difference between the two scores reflects each model's lean toward female or male, which we report as the mean Δ. Finally, across every sentence in our datasets and in the existing GenderLex and WinoBias benchmarks, we compute an agreement score: the percentage of contexts in which the context-based and pronoun-based biases point to the same gender.

The results of this novel evaluation show that the models largely disagree on the direction of bias between context and pronoun, indicating that both signals must be examined bidirectionally to identify the biases embedded in these language models. In addition, action verbs tend to be more male-leaning, whereas object nouns tend to be more female-leaning, suggesting that the models see men as more active agents than women.

Future work could continue research on the bidirectional bias assessment in language models. As with the current thesis's short dataset, the model's predictions were less reliable; future research could examine the effect of sentence context length on the (bidirectional) bias evaluation. In addition, a broader context selection could include research beyond using object nouns, action verbs, and occupations, for example, examining the bias that descriptions of marginalised individuals in society introduce into models.

Furthermore, future research could examine biases across regions. For example, choosing occupation titles common in the European Union or researching the Estonian language and the biases those differences introduce into language models.

In addition, it would be useful to include the gender-neutral pronoun they/them in research on pronoun selection, as it is widely used to refer to someone whose gender is unknown and provides models with another option to signal gender-neutral biases.

8. Acknowledgements

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9. Use of AI

Generative AI in the thesis was mainly used to fix the grammar and wording of the thesis with the University of Tartu Grammarly [16] subscription. Generative AI was used to create the LaTeX tables in the thesis in a visually pleasing format. In addition, the AI autocorrect and coding assistant features in the JetBrains PyCharm and Visual Studio Code IDEs were used to fix parts of the code for the datasets. The agentic workflow for lengthening and creating sentences was partly developed using the GitHub Copilot Education subscription. The author of the thesis made all architectural and structural decisions and verified the code.

[16] More information can be found here https://www.grammarly.com/edu

References

[1] Gallegos I. O., Rossi R. A., Barrow J., Tanjim M. M., Kim S., Dernoncourt F., Yu T., Zhang R., and Ahmed N. K. Bias and Fairness in Large Language Models: A Survey. *Computational Linguistics* 50.3 (Sept. 2024), pp. 1097–1179. DOI: 10.1162/coli_a_00524. https://aclanthology.org/2024.cl-3.8/.

[2] OpenAI. GPT-5 System Card. https://openai.com/research/gpt-5-system-card. Accessed: 2025-09-10. 2025.

[3] Devlin J., Chang M.-W., Lee K., and Toutanova K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *North American Chapter of the Association for Computational Linguistics*. 2019. https://api.semanticscholar.org/CorpusID:52967399.

[4] Warner B., Chaffin A., Clavié B., Weller O., Hallström O., Taghadouini S., Gallagher A., Biswas R., Ladhak F., Aarsen T., Cooper N., Adams G., Howard J., and Poli I. Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference. 2024. DOI: 10.48550/ARXIV.2412.13663. https://arxiv.org/abs/2412.13663.

[5] Nangia N., Vania C., Bhalerao R., and Bowman S. R. CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Ed. by Webber B., Cohn T., He Y., and Liu Y. Online: Association for Computational Linguistics, Nov. 2020, pp. 1953–1967. DOI: 10.18653/v1/2020.emnlp-main.154. https://aclanthology.org/2020.emnlp-main.154/.

[6] Nadeem M., Bethke A., and Reddy S. StereoSet: Measuring stereotypical bias in pretrained language models. *ACL)*. 2021.

[7] Zhao J., Wang T., Yatskar M., Ordonez V., and Chang K.-W. Gender Bias in Coreference Resolution: Evaluation and Debiasing Methods. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*. Ed. by Walker M., Ji H., and Stent A. New Orleans, Louisiana: Association for Computational Linguistics, June 2018, pp. 15–20. DOI: 10.18653/v1/N18-2003. https://aclanthology.org/N18-2003/.

[8] Rudinger R., Naradowsky J., Leonard B., and Van Durme B. Gender Bias in Coreference Resolution. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2*

(Short Papers). Ed. by Walker M., Ji H., and Stent A. New Orleans, Louisiana: Association for Computational Linguistics, June 2018, pp. 8–14. DOI: 10.18653/v1/N18-2002. https://aclanthology.org/N18-2002/.

[9] Sabir A. and Sharma R. Exploring Gender Bias Beyond Occupational Titles. 2025. arXiv: 2507.02679 [cs.CL]. https://arxiv.org/abs/2507.02679.

[10] Eagly A. H. and Wood W. The Nature–Nurture Debates: 25 Years of Challenges in Understanding the Psychology of Gender. *Perspectives on Psychological Science* 8.3 (May 2013), pp. 340–357. DOI: 10.1177/1745691613484767. http://dx.doi.org/10.1177/1745691613484767.

[11] Gender Equality E. I. for. Gender Equality Index 2025. Gender Equality Index Report. LU: Publications Office of the European Union, 2025. DOI: 10.2839/8800867. https://data.europa.eu/doi/10.2839/8800867.

[12] King T. L. and Elliott K. Why Gender Equality Is Good for Men's Health and Why This Matters Now. en. *American Journal of Preventive Medicine* 60.6 (June 2021), pp. 873–876. DOI: 10.1016/j.amepre.2021.01.007. https://linkinghub.elsevier.com/retrieve/pii/S0749379721000866 (04/17/2026).

[13] Kaukonen E., Oskolskaia P., Lindström L., and Marling R. Gender, language and labour: gender perception of Estonian and Russian occupational titles. *Frontiers in Communication* Volume 9 - 2024 (2025). DOI: 10.3389/fcomm.2024.1454022. https://www.frontiersin.org/journals/communication/articles/10.3389/fcomm.2024.1454022.

[14] Hardmeier C., Costa-jussà M. R., Webster K., Radford W., and Blodgett S. L. How to Write a Bias Statement: Recommendations for Submissions to the Workshop on Gender Bias in NLP. 2021. arXiv: 2104.03026 [cs.CL]. https://arxiv.org/abs/2104.03026.

[15] Nadeem M., Bethke A., and Reddy S. StereoSet: Measuring stereotypical bias in pretrained language models. *arXiv preprint arXiv:2004.09456* (2020).

[16] Salazar J., Liang D., Nguyen T. Q., and Kirchhoff K. Masked language model scoring. *Proceedings of the 58th annual meeting of the association for computational linguistics*. 2020, pp. 2699–2712.

[17] Kauf C. and Ivanova A. A Better Way to Do Masked Language Model Scoring. en. arXiv:2305.10588 [cs]. May 2023. DOI: 10.48550/arXiv.2305.10588. http://arxiv.org/abs/2305.10588 (05/05/2026).

[18] Cohen J. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20 (1960), pp. 37–46. https://api.semanticscholar.org/CorpusID:15926286.

[19] Liu Y., Ott M., Goyal N., Du J., Joshi M., Chen D., Levy O., Lewis M., Zettlemoyer L., and Stoyanov V. RoBERTa: A Robustly Optimized BERT Pretraining Approach. 2019. DOI: 10.48550/ARXIV.1907.11692. https://arxiv.org/abs/1907.11692.

[20] Sanh V., Debut L., Chaumond J., and Wolf T. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. 2019. DOI: 10.48550/ARXIV.1910.01108. https://arxiv.org/abs/1910.01108.

[21] Beauvoir S. de. The Second Sex. New York, NY: Vintage Books, Mar. 1991.

[22] Knowles C. Beyond adaptive preferences: Rethinking women's complicity in their own subordination. en. *European Journal of Philosophy* 30.4 (Dec. 2022), pp. 1317–1334. DOI: 10.1111/ejop.12742. https://onlinelibrary.wiley.com/doi/10.1111/ejop.12742 (05/10/2026).

Appendices

You can make separate sub-sections here for Your glossary, description of accompanying files, user manual, large pictures, and tables.

9.1 Prompts

9.1.1 Prompt Dataset Triplets

The following prompt was used to generate the triplets, which included occupation, a related verb and a related noun:

> **Prompt 1:**
>
> You are given ten occupations in alphabetical order as follows:
>
> *the ten occupations are given as a list*
>
> I want you to do the following:
>
> 1. Look through the occupations
>
> 2. For each of the occupations I want you to generate three triplets with the following conditions:
>
> 2.1 The triplets are in the following order (occupation, object noun, action verb).
>
> 2.2 You need to generate 10 unique triplets for each occupations, each object noun and action verb can repeat maximum two times.
>
> 2.3 You need to choose object nouns and action verbs that you think would fit with the occupation.
>
> 3 Once you have generated 10 unique triplets for each occupation, check through all triplets to see if they fulfil the requirements
>
> 4. Give me the triplets as ten python dictionaries, where in each dictionary there are three keys:
>
> 1. A key *occupations* with values of occupations as a python list
>
> 2. A key *object-nouns* with values of object nouns as a python list that index-wise match with the occupations.
>
> 3. A key *actions-verbs* with values of action verbs as a python list that match with object nouns and occupations to index-wise form the correct triplets.

9.1.2 Prompt Short Sentences

The following prompt was used to generate the triplets, which included occupation, a related verb and a related noun:

Prompt 2:

You create GenderLex-style bias-eval pairs. Follow these rules strictly:

1. You will be given explicit (occupation, object-noun, action-verb) assignments.
2. For each assignment produce two sentences: sentence-him and sentence-her.
3. The sentences must be identical except for the final word, which is either 'him' or 'her'.
4. Use neutral, professional language; avoid sensitive or demeaning content.
5. The pronoun must be the final token (optionally followed by punctuation like '.' or '!').
6. Keep sentences single-line and under 35 words.
7. Return JSON that matches the provided schema exactly; no extra commentary.

The following additional prompt was used in addition to the base prompt:

Prompt 3:

Task: Generate n GenderLex instances in the given order. Each must honor the assignment exactly and end with a pronoun cloze.

Assignments (index, occupation, object-noun, action-verb): ...

The following JSON schema 1 was given as an additional context together with the prompts.

9.1.3 Prompts Long Sentence Dataset

The following prompts were used to generate longers sentences from the shorter sentences;

Prompt 4:

You are a sentence editor. You will receive sentences that end with the word 'him'. Your job is to make each sentence longer so that it contains between 11 and 14 words (inclusive). Follow these rules strictly:

1. Keep the original sentence structure and word order intact. Do NOT rearrange any existing words.
2. Only ADD meaningful words or short phrases that provide additional context. Every original word must remain in the sentence, unchanged.
3. The sentence must still end with 'him.' as the last word.
4. The first two words of the original sentence MUST remain together as the first two words of the lengthened sentence. Do NOT insert any words before or between them.
5. Do NOT use special characters such as hyphens (-), apostrophes ('), backticks (‘), or acute accents. Commas are allowed.

```
{
    JSON_SCHEMA = {
  "name": "genderlex_model_generated",
  "schema": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "dataset_name": { "type": "string" },
      "num_instances": { "type": "integer", "minimum": 1 },
      "instances": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": "object",
          "additionalProperties": false,
          "properties": {
            "index": { "type": "integer" },
            "occupation": { "type": "string" },
            "object_noun": { "type": "string" },
            "action_verb": { "type": "string" },
            "sentence_him": { "type": "string" },
            "sentence_her": { "type": "string" }
          },
          "required": [
            "index",
            "occupation",
            "object_noun",
            "action_verb",
            "sentence_him",
            "sentence_her"
          ]
        }
      }
    },
    "required": ["dataset_name", "num_instances", "instances"]
  }
}
}
```

Listing 1. JSON schema for creating the short dataset

6. The output must be a single sentence. Do NOT split it into multiple sentences or add extra periods, exclamation marks, or question marks in the middle.
7. Use neutral, professional language.
8. Return a JSON object with a single key ’sentences’ whose value is a list of strings, one per input sentence, in the same order.
)

Prompt 5:
You are a strict sentence validator. You will receive pairs of (original-sentence, lengthened-sentence). For each pair, check ALL of the following rules and return true only if every rule passes:
1. The lengthened sentence has between 11 and 14 words (inclusive).
2. Every word from the original sentence appears in the lengthened sentence in the same relative order.
3. The lengthened sentence ends with ’him.’ as the final word.
4. The first two words of the original sentence remain the first two words of the lengthened sentence (nothing inserted before or between them).
5. There are no special characters such as hyphens (-), apostrophes (’), backticks (‘), or acute accents.
6. There is only one sentence (no extra periods, exclamation marks, or question marks in the middle).
7. The added words are meaningful and make grammatical sense.
Return a JSON object with a single key ’results’ whose value is a list of booleans, one per pair, in the same order.

9.2 PLL Original and Word l2r Comparison

Here are displayed all the aggregated comparison scores for all of the datasets between $PLL(w)_{l2r}$ (within_word) and $PLL(w)$ (original) scores.

Table 10. Comparison of $PLL(w)$ and $PLL(w)_{l2r}$ scores for WinoBias. Statistically significant differences are marked with the $PLL(w)_{l2r}$ scores and statistical significance is indicated by stars: $^{*}p \leq 0.05$, $^{**}p \leq 0.01$, $^{***}p \leq 0.001$.

		Context Masking						Pronoun Masking		
		Action Verb (%)			Occupation (%)			All (%)		
Model	Method	M	F	mean	M	F	mean	M	F	mean
BERT	$PLL(w)$	51.1	48.9	0.009	75.4	24.6	0.690	79.1	20.9	1.796
	$PLL(w)_{l2r}$	50.8	49.2	0.006	74.1	25.9	0.721^{**}	79.1	20.9	1.796
RoBERTa	$PLL(w)$	52.5	47.5	0.020	58.7	41.3	0.253	73.9	26.1	0.904
	$PLL(w)_{l2r}$	52.7	47.3	0.018	60.2	39.8	0.292^{**}	73.9	26.1	0.904
ModernBERT	$PLL(w)$	49.8	50.2	-0.001	61.3	38.7	0.237	76.8	23.2	0.865
	$PLL(w)_{l2r}$	49.7	50.3	-0.002	61.9	38.1	0.224^{**}	76.8	23.2	0.865
DistilBERT	$PLL(w)$	62.5	37.5	0.070	69.5	30.5	0.182	91.5	8.5	1.743
	$PLL(w)_{l2r}$	62.2	37.8	0.072	69.2	30.8	0.170^{***}	91.5	8.5	1.743
DistilRoBERTa	$PLL(w)$	51.5	48.5	-0.022	61.8	38.2	0.263	80.9	19.1	0.669
	$PLL(w)_{l2r}$	51.9	48.1	-0.026	61.7	38.3	0.271	80.9	19.1	0.669

Table 11. Comparison of $PLL(w)$ and $PLL(w)_{l2r}$ scores for Winogender. Statistically significant differences marked with the $PLL(w)_{l2r}$ scores and statistical significance is indicated by stars: $^{*}p \leq 0.05$, $^{**}p \leq 0.01$, $^{***}p \leq 0.001$.

		Context Masking						Pronoun Masking		
		Action Verb (%)			Occupation (%)			All (%)		
Model	Method	M	F	mean	M	F	mean	M	F	mean
BERT	$PLL(w)$	50.6	49.4	0.024	70.2	29.8	0.328	89.9	10.1	2.310
	$PLL(w)_{l2r}$	50.6	49.4	0.029	73.0	27.0	0.405^{*}	89.9	10.1	2.310
RoBERTa	$PLL(w)$	49.4	50.6	0.005	59.0	41.0	0.067	73.3	26.7	0.581
	$PLL(w)_{l2r}$	50.6	49.4	0.011	65.7	34.3	0.096	73.3	26.7	0.581
ModernBERT	$PLL(w)$	52.2	47.8	0.016	58.4	41.6	0.079	86.8	13.2	0.945
	$PLL(w)_{l2r}$	52.2	47.8	0.016	60.2	39.8	0.077	86.8	13.2	0.945
DistilBERT	$PLL(w)$	55.6	44.4	0.010	56.2	43.8	0.070	89.9	10.1	1.018
	$PLL(w)_{l2r}$	56.7	43.3	0.015^{*}	53.9	46.1	0.038^{*}	89.9	10.1	1.018
DistilRoBERTa	$PLL(w)$	46.6	53.4	0.007	38.0	62.0	-0.102	32.3	67.7	-0.473
	$PLL(w)_{l2r}$	47.8	52.2	0.008	38.5	61.5	-0.103	32.3	67.7	-0.473

Table 12. Comparison of $PLL(w)$ and $PLL(w)_{l2r}$ scores for GenderLex. Statistically significant differences marked with the $PLL(w)_{l2r}$ scores and statistical significance is indicated by stars: $^{*}p \leq 0.05$, $^{**}p \leq 0.01$, $^{***}p \leq 0.001$.

		Context Masking									Pronoun Masking		
		Action Verb (%)			Object Noun (%)			Occupation (%)			All (%)		
Model	Method	M	F	mean	M	F	mean	M	F	mean	M	F	mean
BERT	$PLL(w)$	66.9	33.1	0.149	48.6	51.4	0.005	62.7	37.3	0.132	67.9	32.1	0.211
	$PLL(w)_{l2r}$	66.5	33.5	0.147	49.7	50.3	0.011^{*}	63.9	36.1	0.156^{**}	67.9	32.1	0.211
RoBERTa	$PLL(w)$	51.1	48.9	-0.017	49.8	50.2	-0.005	62.5	37.5	0.253	79.0	21.0	0.912
	$PLL(w)_{l2r}$	50.4	49.6	-0.026^{**}	49.9	50.1	-0.006	61.4	38.6	0.231	79.0	21.0	0.912
ModernBERT	$PLL(w)$	78.8	21.2	0.344	52.2	47.8	0.012	80.6	19.4	0.415	27.0	73.0	-0.634
	$PLL(w)_{l2r}$	78.8	21.2	0.344	52.1	47.9	0.011	80.8	19.2	0.415	27.0	73.0	-0.634
DistilBERT	$PLL(w)$	87.7	12.3	0.624	48.6	51.4	-0.019	65.0	35.0	0.136	89.2	10.8	1.156
	$PLL(w)_{l2r}$	87.8	12.2	0.628	48.3	51.7	-0.019	68.2	31.8	0.152^{***}	89.2	10.8	1.156
DistilRoBERTa	$PLL(w)$	79.1	20.9	0.267	52.9	47.1	0.010	60.9	39.1	0.133	87.8	12.2	0.947
	$PLL(w)_{l2r}$	79.1	20.9	0.275^{*}	53.2	46.8	0.010	58.6	41.4	0.116^{***}	87.8	12.2	0.947

Table 13. Comparison of $PLL(w)$ and $PLL(w)_{l2r}$ scores for thesis's short dataset. Statistically significant differences marked together with the $PLL(w)_{l2r}$ scores and statistical significance is indicated by stars: $^{*}p \leq 0.05$, $^{**}p \leq 0.01$, $^{***}p \leq 0.001$.

		Context Masking									Pronoun Masking		
		Action Verb (%)			Object Noun (%)			Occupation (%)			All (%)		
Model	Method	M	F	mean	M	F	mean	M	F	mean	M	F	mean
BERT	$PLL(w)$	39.8	60.2	-0.208	32.0	68.0	-0.683	37.8	62.2	-0.097	28.5	71.5	-0.395
	$PLL(w)_{l2r}$	40.2	59.8	-0.192	34.8	65.2	-0.536^{**}	41.5	58.5	-0.113	28.5	71.5	-0.395
RoBERTa	$PLL(w)$	35.0	65.0	-0.189	38.0	62.0	-0.126	43.2	56.8	-0.115	75.0	25.0	0.680
	$PLL(w)_{l2r}$	34.0	66.0	-0.252^{***}	37.8	62.2	-0.156^{*}	45.5	54.5	-0.122	75.0	25.0	0.680
ModernBERT	$PLL(w)$	40.2	59.8	-0.125	45.0	55.0	-0.065	49.0	51.0	0.009	48.8	51.2	-0.083
	$PLL(w)_{l2r}$	40.2	59.8	-0.126	45.0	55.0	-0.065	48.5	51.5	-0.004^{*}	48.8	51.2	-0.083
DistilBERT	$PLL(w)$	54.8	45.2	0.071	42.2	57.8	-0.040	35.8	64.2	-0.153	68.8	31.2	0.495
	$PLL(w)_{l2r}$	55.8	44.2	0.086^{*}	41.5	58.5	-0.052	35.2	64.8	-0.181^{***}	68.8	31.2	0.495
DistilRoBERTa	$PLL(w)$	52.2	47.8	-0.008	54.2	45.8	0.027	39.0	61.0	-0.252	54.2	45.8	0.040
	$PLL(w)_{l2r}$	50.8	49.2	-0.021^{*}	55.2	44.8	0.030	35.5	64.5	-0.279^{*}	54.2	45.8	0.040

Table 14. Comparison of $PLL(w)$ and $PLL(w)_{l2r}$ scores for thesis's long dataset. Statistically significant differences marked together with the $PLL(w)_{l2r}$ scores and statistical significance is indicated by stars: $^{*}p \leq 0.05$, $^{**}p \leq 0.01$, $^{***}p \leq 0.001$.

		Context Masking									Pronoun Masking		
		Action Verb (%)			Object Noun (%)			Occupation (%)			All (%)		
Model	Method	M	F	mean	M	F	mean	M	F	mean	M	F	mean
BERT	$PLL(w)$	41.0	59.0	-0.071	48.8	51.2	-0.003	48.8	51.2	-0.016	33.5	66.5	-0.364
	$PLL(w)_{l2r}$	42.2	57.8	-0.053^{*}	49.5	50.5	-0.010	52.5	47.5	-0.006	33.5	66.5	-0.364
RoBERTa	$PLL(w)$	35.0	65.0	-0.158	38.5	61.5	-0.102	47.0	53.0	-0.038	53.2	46.8	0.075
	$PLL(w)_{l2r}$	33.8	66.2	-0.186^{*}	40.8	59.2	-0.105	49.2	50.8	-0.059	53.2	46.8	0.075
ModernBERT	$PLL(w)$	40.5	59.5	-0.089	44.0	56.0	-0.068	57.5	42.5	0.116	28.8	71.2	-0.767
	$PLL(w)_{l2r}$	40.5	59.5	-0.091	43.8	56.2	-0.070	58.2	41.8	0.116	28.8	71.2	-0.767
DistilBERT	$PLL(w)$	58.2	41.8	0.043	43.5	56.5	-0.042	49.8	50.2	-0.053	73.8	26.2	0.592
	$PLL(w)_{l2r}$	58.2	41.8	0.049	45.0	55.0	-0.041	47.2	52.8	-0.082^{***}	73.8	26.2	0.592
DistilRoBERTa	$PLL(w)$	45.2	54.8	-0.012	48.2	51.8	-0.011	41.8	58.2	-0.132	47.8	52.2	-0.128
	$PLL(w)_{l2r}$	44.0	56.0	-0.016	48.8	51.2	-0.007	39.2	60.8	-0.161^{*}	47.8	52.2	-0.128

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